Typical Satellite Encounter Scenarios
for ORbital Architectures for Cooperative Laser Energetics (ORACLE) Project

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I. Introduction

The aim of this report is to provide seven typical satellite encounter cases that might happen in a multi-shell open-cavity laser-interlinked constellation. In all of these test cases, there will be multiple helper satellites evenly distributed on a lower equatorial circular orbit at 1000 km and a target satellite placed on a higher or inclined circular orbit, as shown in Fig. 1.

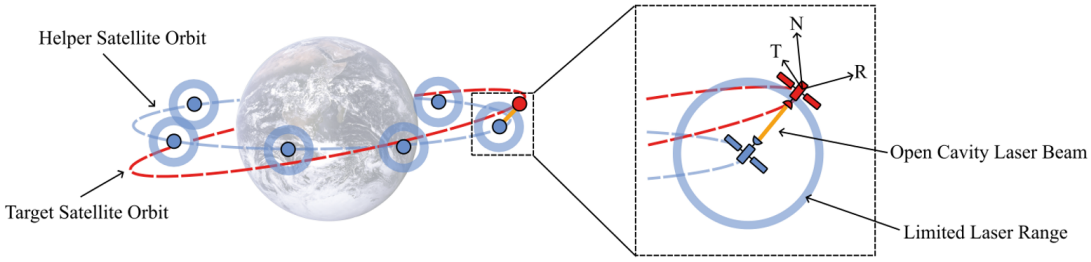


Fig. 1 Illustration of general test case geometry.

The laser interlink parameters used are summarized below in Table 1.

Table 1 Laser interlink parameters used.

Parameter	Symbol / Setting	Value
Input laser power	P_{in}	10 kW
Cavity power-buildup factor	B	100
Maximum laser-link range	$L_{\max}$	200 km
Geometric transmission factor	β	1
Spacecraft mass	m	227 kg
Nominal force per active link	$F = \beta B P_{\text{in}}/c$	$\approx$ 3.34 mN

An encounter happens when two satellites come into range and are in line-of-sight. A laser is fired during an encounter according to the schedulers' decision. The two schedulers used in this report are summarized in Table 2. For most test cases, the Semi-major Axis Scheduler is applied. However, in Test Case 5 in Section IV.B, the Inclination Scheduler is applied to show the effect of changing scheduler.

Table 2 Schedulers used in this report.

Scheduler	Objective
Semi-major Axis Raising	Selects the best helper to increase the target's semimajor axis.
Inclination Raising	Selects the best helper to increase the target's orbital inclination.

The orbital parameters chosen for the seven test case are shown in Table 3. These test cases will be presented in three scenario groups. In Section III, the co-planar scenario group will be presented with the target satellite placed on a higher

equatorial orbit. Here, the effect of shell separation distance and number of helpers on encounter opportunities will be shown. In Sections IV, the inclined scenario group will be presented with the target satellite placed on an inclined orbit. Here, the effect of different laser-link scheduler will also be shown. In Section V, the co-orbit scenario group will be presented with the target satellite starting on the same orbit as the helper satellite.

Table 3 Orbital parameters for test cases

Test Case	Target			Helper			Scheduler
	Altitude (km)	Inclination (°)	Initial ν (°)	Altitude (km)	Inclination (°)	Amount	
1	1050	0	0	1000	0	20	Semi-major Axis Raising
2	1150	0	0	1000	0	20	Semi-major Axis Raising
3	1050	0	0	1000	0	100	Semi-major Axis Raising
4	1000	5	0	1000	0	20	Semi-major Axis Raising
5	1000	5	0	1000	0	20	Inclination Raising
6	1050	5	0	1000	0	20	Semi-major Axis Raising
7	1000	0	1	1000	0	20	Semi-major Axis Raising

II. Output Format

For each test case, the satellites are propagated for approximately 50 orbits, and the following analyses are provided in CSV format:

- Encounter and Laser-On Records
- Extrema Analysis
- Maximum-Duration Encounter Trajectory

Here, the Encounter Records contain information for each encounter that occurs during the simulation, including (1) the IDs of the two satellites involved, (2) the start and end times of the encounter, and (3) the encounter duration. The Laser-On Records contain information for each laser-firing event that occurs during the simulation, including (1) the encounter during which the laser is fired, (2) the start and end times of the laser firing, and (3) the laser-firing duration.

The Extrema Analysis reports the minimum and maximum values of (1) relative speed, (2) range rate, (3) encounter duration, and (4) laser-on duration, observed over the entire simulation. This record also contains the encounter ID at which each extremum occurs.

For each simulation, the states of the two satellites involved in the maximum-duration encounter are recorded in Maximum-Duration Encounter Trajectory.csv. The states are sampled at 10-second intervals and recorded in Earth-centered inertial (ECI) Cartesian coordinates. The information contained in each column of this CSV file is summarized in Table 4.

Table 4 Columns in the Maximum-Duration Encounter Trajectory CSV file

Column(s)	**Description**	**Unit**
`spacecraft_a`	ID of spacecraft A	None
`spacecraft_b`	ID of spacecraft B	None
`time_s`	Elapsed time since simulation start	s
`spacecraft_a_r_[x,y,z]_m`	Spacecraft A's ECI position components	m
`spacecraft_a_v_[x,y,z]_m_s`	Spacecraft A's ECI velocity components	m/s
`spacecraft_b_r_[x,y,z]_m`	Spacecraft B's ECI position components	m
`spacecraft_b_v_[x,y,z]_m_s`	Spacecraft B's ECI velocity components	m/s
`laser_on`	Laser fired at `time_s` (`true` = on, `false` = off)	Boolean

III. Scenario Group A: Co-planar Altitude-Separated

A. Test Case 1: Moderate Separation

1. Setup

In this test case, the target is placed at a slightly higher orbit than the helper satellite orbit, as shown in Fig. 2. The orbital parameters and scheduler used for this case are recorded in Table 3. This case is meant to be the baseline case for other test cases to compare with.

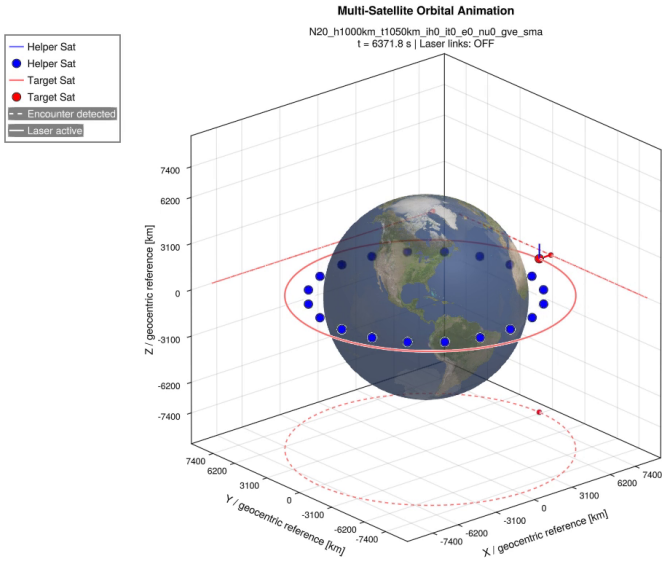


Fig. 2 Test Case 1 Setup [Click Here for Animation].

2. Results

In this test case 1, 11 encounters are detected, as recorded in [Here], and 11 laser firings are activated, as recorded in [Here]. The extrema analysis for this test case is shown in Table 5. The states of the two satellites in the maximum-duration encounter (encounter 11) are recorded in [Here]. For verification, the simulation was also run on a low-fidelity model, and the result comparison can be found in [Here].

Table 5 Extrema analysis for Test Case 1

Metric	Extremum	Value	Start Time	Encounter ID
Encounter speed	min	16.40 m/s	309700.00 s	11
	max	211.29 m/s	275720.93 s	10
Range rate	min	−92.70 m/s	306592.57 s	11
	max	103.74 m/s	311900.00 s	11
Encounter duration	min	2591.86 s	0.00 s	1
	max	5367.86 s	306592.57 s	11
Laser-on duration	min	0.00 s	0.00 s	1
	max	3422.34 s	275720.93 s	10

B. Test Case 2: Increased Separation

1. Setup

In this test case, the target is placed at a significantly higher orbit than the helper satellite orbit, as shown in Fig. 3. The orbital parameters and scheduler used for this case are recorded in Table 3.

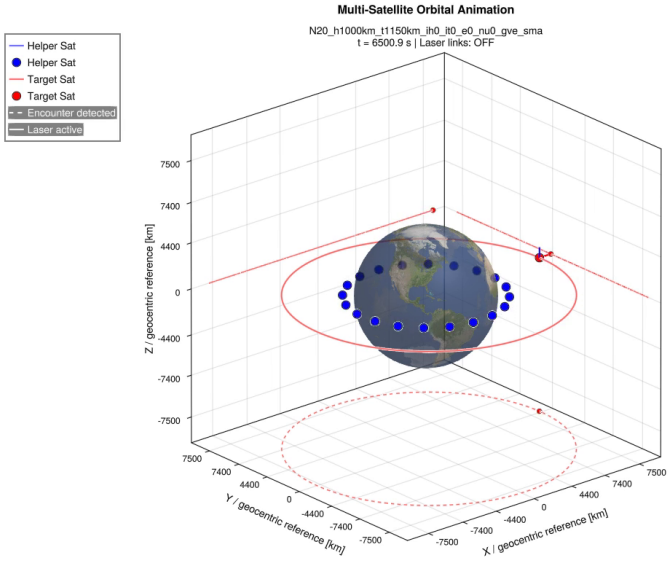


Fig. 3 Test Case 2 Setup [Click Here for Animation].

2. Results

In this test case 2, 31 encounters are detected, as recorded in [Here], and 31 laser firings are activated, as recorded in [Here]. The extrema analysis for this test case is shown in Table 6. The states of the two satellites in the maximum-duration encounter (encounter 13) are recorded in [Here]. For verification, the simulation was also run on a low-fidelity model, and the result comparison can be found in [Here].

Table 6 Extrema analysis for Test Case 2

Metric	Extremum	Value	Start Time	Encounter ID
Encounter speed	min	57.91 m/s	126530.00 s	13
	max	166.66 m/s	127282.78 s	13
Range rate	min	−158.22 m/s	294340.75 s	29
	max	158.72 m/s	316877.73 s	31
Encounter duration	min	598.33 s	0.00 s	1
	max	1503.27 s	125779.50 s	13
Laser-on duration	min	0.00 s	0.00 s	1
	max	788.32 s	94200.27 s	10

C. Test Case 3: Increased Helper Count

1. Setup

In this test case, the helper number is increased to 100, as shown in Fig. 4. The orbital parameters and scheduler used for this case are recorded in Table 3.

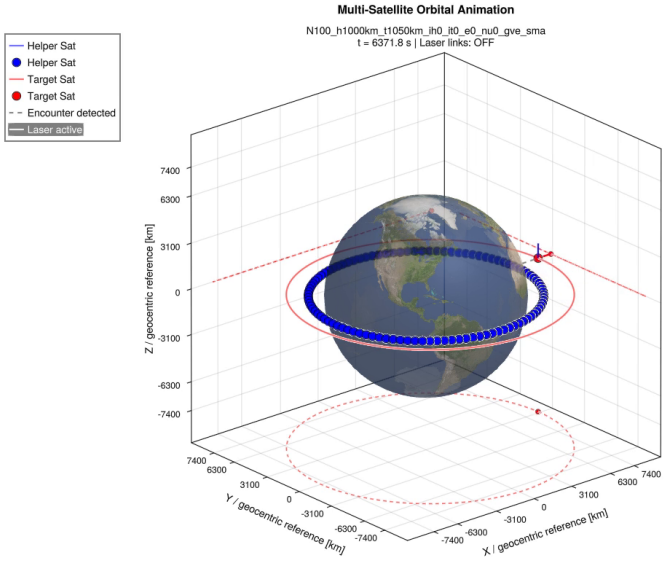


Fig. 4 Test Case 3 Setup [Click Here for Animation].

2. Results

In this test case 3, 54 encounters are detected, as recorded in [Here], and 54 laser firings are activated, as recorded in [Here]. The extrema analysis for this test case is shown in Table 7. The states of the two satellites in the maximum-duration encounter (encounter 36) are recorded in [Here]. For verification, the simulation was also run on a low-fidelity model, and the result comparison can be found in [Here].

Table 7 Extrema analysis for Test Case 3

Metric	Extremum	Value	Start Time	Encounter ID
Encounter speed	min	8.52 m/s	308520.00 s	52
	max	211.37 m/s	282063.97 s	48
Range rate	min	−108.45 m/s	312010.38 s	53
	max	108.38 m/s	305196.53 s	51
Encounter duration	min	538.54 s	318028.50 s	54
	max	5271.40 s	210978.99 s	36
Laser-on duration	min	0.00 s	0.00 s	1
	max	3362.35 s	287986.68 s	49

IV. Scenario Group B: Inclined Target Orbit

A. Test Case 4: Semi-major Axis Scheduler

1. Setup

In this test case, the target is placed on a 5 deg inclined orbit with the same altitude as the helper orbit, as shown in Fig. 5. The Semi-major Axis Scheduler is used to increase target's semi-major axis. The orbital parameters and scheduler used for this case are recorded in Table 3.

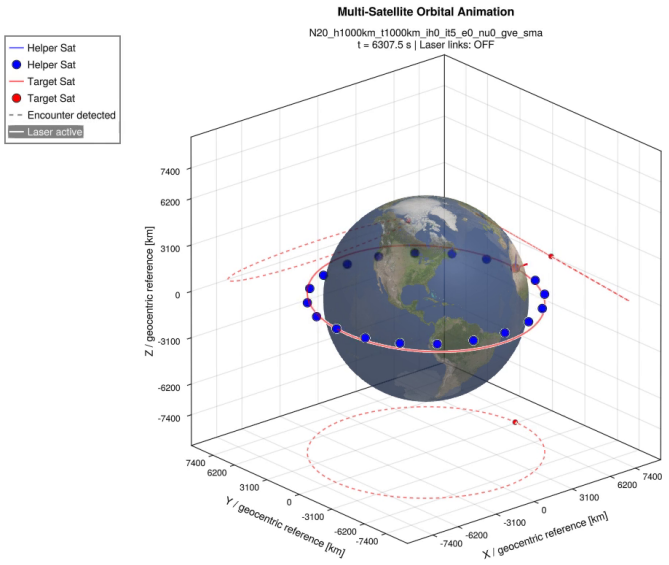


Fig. 5 Test Case 4 Setup [Click Here for Animation].

2. Results

In this test case 4, 101 encounters are detected, as recorded in [Here], and 20 laser firings are activated, as recorded in [Here]. The extrema analysis for this test case is shown in Table 8. The states of the two satellites in the maximum-duration encounter (encounter 3) are recorded in [Here]. For verification, the simulation was also run on a low-fidelity model, and the result comparison can be found in [Here].

Table 8 Extrema analysis for Test Case 4

Metric	Extremum	Value	Start Time	Encounter ID
Encounter speed	min	609.37 m/s	317.18 s	1
	max	643.93 m/s	311070.00 s	100
Range rate	min	−642.33 m/s	3120.00 s	2
	max	642.34 m/s	3170.00 s	2
Encounter duration	min	317.18 s	0.00 s	1
	max	634.33 s	5967.07 s	3
Laser-on duration	min	0.00 s	62525.57 s	21
	max	317.17 s	0.00 s	1

B. Test Case 5: Inclination Scheduler

1. Setup

In this test case, the target is placed on a 5 deg inclined orbit with the same altitude as the helper orbit, as shown in Fig. 6. The Inclination Scheduler is used to raise target's orbit inclination in comparison with Test Case 4. The orbital parameters and scheduler used for this case are recorded in Table 3.

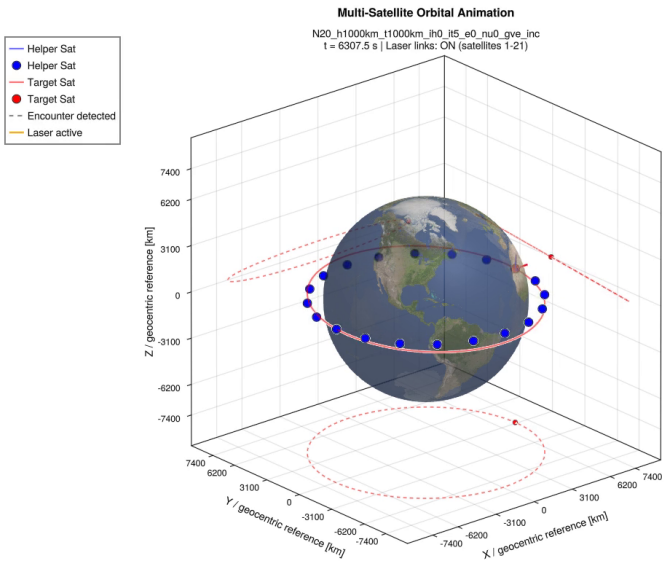


Fig. 6 Test Case 5 Setup [Click Here for Animation].

2. Results

In this test case 5, 101 encounters are detected, as recorded in [Here], and 101 laser firings are activated, as recorded in [Here]. The extrema analysis for this test case is shown in Table 9. The states of the two satellites in the maximum-duration encounter (encounter 3) are recorded in [Here]. For verification, the simulation was also run on a low-fidelity model, and the result comparison can be found in [Here].

Table 9 Extrema analysis for Test Case 5

Metric	Extremum	Value	Start Time	Encounter ID
Encounter speed	min	609.37 m/s	317.18 s	1
	max	643.45 m/s	311070.00 s	100
Range rate	min	−643.07 m/s	267060.00 s	86
	max	643.06 m/s	267110.00 s	86
Encounter duration	min	317.18 s	0.00 s	1
	max	634.33 s	5967.07 s	3
Laser-on duration	min	308.68 s	141081.55 s	46
	max	317.17 s	0.00 s	1

C. Test Case 6: Inclination Target Orbit With Separation

1. Setup

In this test case, the target is placed on a 5 deg inclined orbit with the 50 km altitude separation with the helper orbit, as shown in Fig. 7. The orbital parameters and scheduler used for this case are recorded in Table 3.

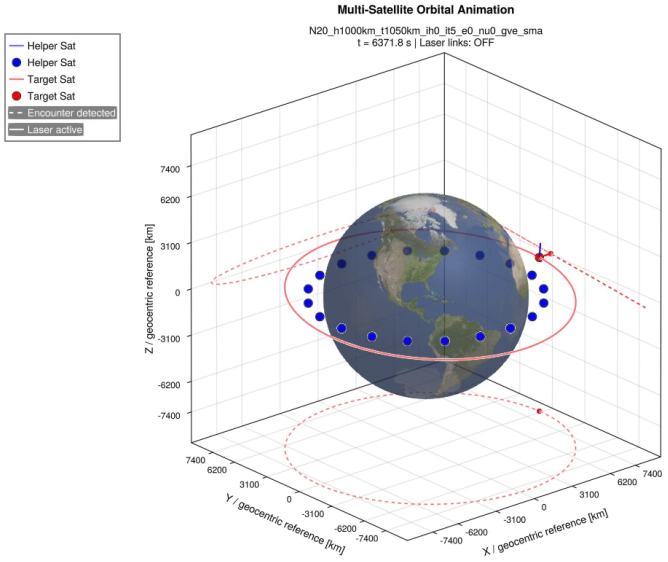


Fig. 7 Test Case 6 Setup [Click Here for Animation].

2. Results

In this test case 6, 18 encounters are detected, as recorded in [Here], and 10 laser firings are activated, as recorded in [Here]. The extrema analysis for this test case is shown in Table 10. The states of the two satellites in the maximum-duration encounter (encounter 8) are recorded in [Here]. For verification, the simulation was also run on a low-fidelity model, and the result comparison can be found in [Here].

Table 10 Extrema analysis for Test Case 6

Metric	Extremum	Value	Start Time	Encounter ID
Encounter speed	min	611.75 m/s	124108.40 s	8
	max	670.57 m/s	127000.00 s	9
Range rate	min	−601.09 m/s	123530.00 s	8
	max	598.90 m/s	124070.00 s	8
Encounter duration	min	160.80 s	126841.31 s	9
	max	611.14 s	123497.26 s	8
Laser-on duration	min	0.00 s	310778.38 s	18
	max	584.29 s	279050.81 s	16

V. Scenario Group C: Co-Orbital Satellites

A. Test Case 7: Co-Orbital Initial Condition

1. Setup

In this test case, the target is placed on the same initial orbit as the helper satellites, but with a small true-anomaly deviation to avoid the target satellite overlapping with another helper satellite. This is shown in Fig. 8. The orbital parameters and scheduler used for this case are recorded in Table 3.

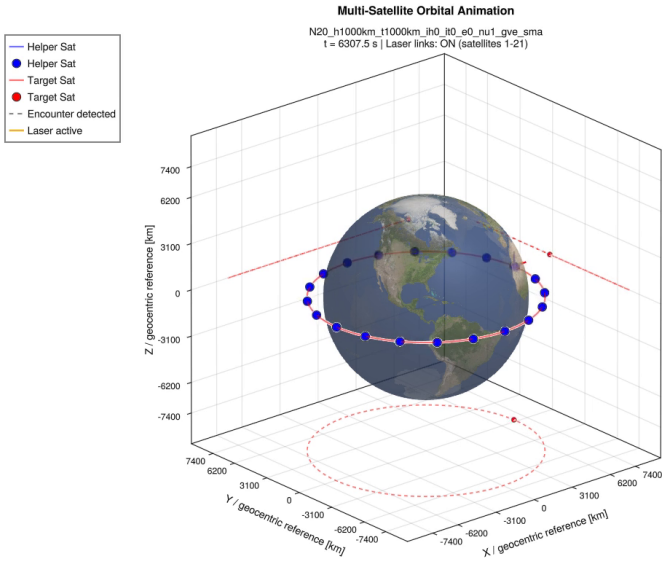


Fig. 8 Test Case 7 Setup [Click Here for Animation].

2. Results

In this test case 7, only 1 encounter was detected, as recorded in [Here], and only 1 laser firing was activated, as recorded in [Here]. The extrema analysis for this test case is shown in Table 11. The states of the two satellites in the maximum-duration encounter (encounter 1) are recorded in [Here]. For verification, the simulation was also run on a low-fidelity model, and the result comparison can be found in [Here].

Table 11 Extrema analysis for Test Case 7

Metric	Extremum	Value	Start Time	Encounter ID
Encounter speed	min	1.59 m/s	53910.00 s	1
	max	199.79 m/s	96362.70 s	1
Range rate	min	−4.53 m/s	51640.00 s	1
	max	5.01 m/s	96260.00 s	1
Encounter duration	min	96362.70 s	0.00 s	1
	max	96362.70 s	0.00 s	1
Laser-on duration	min	53904.20 s	0.00 s	1
	max	53904.20 s	0.00 s	1

VI. Cross-case Comparison

The extrema analysis for all the simulations done across 7 test cases is shown in Table 12.

Table 12 Cross-case extrema analysis across all test cases

Metric	Extremum	Value	Encounter ID	Test Case
Encounter speed	min	1.59 m/s	1	7
	max	670.57 m/s	9	6
Range rate	min	−643.07 m/s	86	5
	max	643.06 m/s	86	5
Encounter duration	min	160.80 s	9	6
	max	96362.70 s	1	7
Laser-on duration	min	0.00 s	21	4
	min	0.00 s	18	6
	max	53904.20 s	1	7

VII. Conclusion

In this report, 7 test cases are presented for laser satellite encounter scenarios. Analysis of the extrema is also provided. Code for this report can be found in [Here]. For more questions, please contact kuangsun@umich.edu.